

Comprehensive Presentation Video Script and Screen Guide (Google Colab Version)

Course: MCA 521-4 Machine Learning (CHRIST Deemed to be University)
Assignment: CIA-3 Machine Learning Presentation Video (3 Minutes)
Domain: Mission / Crisis / Disaster Response
Presentation Environment: 100% Google Colab Notebook ([Disaster_Response_Colab.ipynb](#))
Data Verification: 100% exact numerical match with [model_comparison_table.csv](#) and Google Colab execution output.

Part 1: Quick Recording Setup & Section Map

1. Screen Setup

- Open your browser and navigate to your **Google Colab Notebook** ([\[notebooks/Disaster_Response_Colab.ipynb\]](#) (file:///d:/4TH_TREMS/ML/CIA3/notebooks/Disaster_Response_Colab.ipynb)).
- Ensure all cells are executed so that all formatted tables, countplots, word clouds, comparison charts, confusion matrices, ROC curves, live predictions, and SHAP plots are visible.
- Present **entirely from this single Google Colab tab**.

2. Complete Section & Subsection Navigation Map

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TIMELINE	SECTION & SUBSECTION TITLE	CELL TO SHOW IN
GOOGLE COLAB		
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0:00 - 0:20	Section 1: Environment Setup & Packages	Title Cell & Code
Cell 1 (!pip install)		
0:20 - 0:35	Section 1.1: Ethics & Responsible AI Statement	Markdown Cell 1.1
(Privacy, Bias, Human-in-the-loop)		
0:35 - 0:50	Section 2: Dataset Load & Leakage-Safe Preprocess	CSV Load & Clean
Function Code Cells		
0:50 - 1:00	Section 3.1: Target Class Distribution Countplot	Target Bar Chart
(14,288 Related vs 4,828 Not)		
1:00 - 1:10	Section 3.2: Dual Red & Green Word Clouds	Red Emergency
Cloud vs Green General Cloud		
1:10 - 1:25	Section 4: TF-IDF Feature Extraction	TfidfVectorizer
Code Cell (10,000 n-grams)		
1:25 - 1:55	Section 5: Model Selection, 5-Fold CV & Metrics	Results Table
(LR, RF, SVM, XGB, Voting Ensemble)		
1:55 - 2:05	Section 6.1: Grouped Metric Comparison Bar Chart	Multi-Metric Bar
Chart (F1, Accuracy, Precision)		
2:05 - 2:15	Section 6.2 & 6.3: Confusion Matrices & ROC	Grid Heatmaps &
ROC Curves		

2:15 - 2:35	Section 7.1: Live Single-Record Prediction	Synthetic Text Classification Cell (95%+ Confidence)
2:35 - 2:50	Section 7.2 & 7.3: SHAP Feature Importance & Beeswarm Global Feature Bar Chart & Beeswarm Plot	
2:50 - 3:00	Section 8: Model Persistence & Pipeline Artifacts	joblib.dump Code Cell & Closing Remarks
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Part 2: Detailed Subsection-by-Subsection Spoken Video Script

● Section 1: Title & Environment Setup (0:00 – 0:20)

- **[ON-SCREEN ACTION]:** Start recording at the top of your Colab Notebook. Point your cursor at the Title Markdown Cell and Code Cell 1 (`!pip install`).
- **[SPOKEN SCRIPT]:**

*"Hello everyone and respected Professor! Today, I am presenting our CIA-3 Machine Learning project: **CrisisRelief AI: Real-Time Disaster Response Message Classification and Triage System**.*

*We selected the **Crisis Mission Domain**. During major natural disasters—like the 2010 Haiti Earthquake and Pakistan Floods—emergency agencies receive tens of thousands of urgent text messages. Manually reading each message creates critical delays in saving lives.*

*Here in **Section 1**, we set up our environment and install key dependencies including `scikit-learn`, `xgboost`, `shap`, `wordcloud`, and `joblib`."*

● Section 1.1: Ethics and Responsible AI Statement (0:20 – 0:35)

- **[ON-SCREEN ACTION]:** Scroll down slightly to point at the **Section 1.1 Ethics and Responsible AI Statement** Markdown cell.
- **[SPOKEN SCRIPT]:**

*'Section 1.1 outlines our **Ethics and Responsible AI Statement**.*

**Disaster response ML involves life-saving decisions, so we enforce four key ethical principles:*

1. **Privacy Preservation:** All messages were scrubbed of PII, phone numbers, and names prior to analysis.
2. **Demographic Bias Mitigation:** Preprocessing treats direct SMS and news sources objectively across dialects.
3. **Human-in-the-Loop Triage:** AI predictions serve strictly as decision-support indicators for responders; no emergency aid is denied automatically.
4. **Model Auditability:** Using SHAP feature attributions for full algorithmic transparency."*

● Section 2: Dataset Loading & Leakage-Safe Preprocessing (0:35 – 0:50)

- **[ON-SCREEN ACTION]:** Scroll to **Section 2**. Point mouse at the dataset shapes output (**Train: 19116, Val: 2337, Test: 2397**) and the `clean_text` function code cell.
- **[SPOKEN SCRIPT]:**

*"In **Section 2**, we load 25,000+ messages from the **Robert Munro Disaster Response Dataset**.*

*To ensure a **leakage-safe preprocessing pipeline**, all feature transformations were fitted strictly on the training split, keeping validation and test sets untouched.*

Our `clean_text` function normalizes raw text by removing URLs, digits, punctuation, and converting to lowercase, while dropping non-informative zero-variance columns like PII and `child_alone`."

● Section 3.1: Target Class Distribution Countplot (0:50 – 1:00)

- **[ON-SCREEN ACTION]:** Scroll to **Section 3.1** showing the green and red target countplot bar chart.
- **[SPOKEN SCRIPT]:**

*"In **Section 3.1**, our Exploratory Data Analysis shows the target class distribution. Our training dataset contains 14,288 disaster-related emergency messages and 4,828 non-disaster messages. Class imbalance was addressed using balanced class weighting during model training."*

● Section 3.2: Dual Red & Green Word Cloud Comparison (1:00 – 1:10)

- **[ON-SCREEN ACTION]:** Point mouse at the **Section 3.2 Dual Word Clouds** (Red on left, Green on right).
- **[SPOKEN SCRIPT]:**

*"Here in **Section 3.2** are our **Dual Word Clouds**. The red word cloud on the left highlights critical emergency trigger terms such as `water`, `food`, `shelter`, `help`, and `injured`. The green word cloud on the right shows background conversational text."*

● Section 4: TF-IDF Feature Extraction (1:10 – 1:25)

- **[ON-SCREEN ACTION]:** Scroll to **Section 4** and highlight the `TfidfVectorizer` code cell and output matrix shape (`(19116, 10000)`).
- **[SPOKEN SCRIPT]:**

*"In **Section 4**, we perform **TF-IDF Feature Extraction**.*

*We extract **10,000 sublinear TF-IDF unigram and bigram features**. Sublinear TF scaling ($\$1 + \log(\text{tf})\$$) dampens high-frequency word counts, preventing common terms from overwhelming rare emergency words like `trapped` or `hospitals`."*

● Section 5: Model Selection, Training & 5-Fold Cross-Validation (1:25 – 1:55)

- **[ON-SCREEN ACTION]:** Scroll to **Section 5** and highlight the **Final Results Table** row-by-row.
- **[SPOKEN SCRIPT]:**

*"In **Section 5**, we benchmark five classifiers using Stratified 5-Fold Cross Validation on untouched test data:*

1. Our **Baseline Model**—Logistic Regression—achieved a CV F1 of **0.8001** and Test F1 of **0.8054**.
2. Our **Bagging Model (a)**—Random Forest—achieved a CV F1 of **0.7582** and Test F1 of **0.7649**.
3. Our **Linear Classifier**—Linear SVM—achieved a Test F1 of **0.8474**.
4. Our **Boosting Model (b)**—XGBoost—achieved a CV F1 of **0.7532** and Test F1 of **0.8403** (Test Accuracy **85.73%**).
5. Finally, our **Heterogeneous Soft Voting Ensemble (c)** combined LR, RF, SVM, and XGBoost.

*As shown in the table, our **Voting Ensemble outperformed all baseline and individual models**, achieving the highest overall **Test F1-Score of 0.8609 (86.1%)**, **CV F1 of 0.8105**, **Test Accuracy of 85.65%**, **Test Precision of 0.8669**, and **Test ROC-AUC of 0.8888**."*

● Section 6.1: Grouped Metric Comparison Bar Chart (1:55 – 2:05)

- **[ON-SCREEN ACTION]:** Scroll to **Section 6.1** showing the multi-colored grouped bar chart.
- **[SPOKEN SCRIPT]:**

*"In **Section 6.1**, our grouped metric bar chart visually highlights performance across Accuracy, Precision, Recall, and F1-Score, showing clear dominance by our Voting Ensemble."*

● Section 6.2 & 6.3: Confusion Matrices & ROC Curves (2:05 – 2:15)

- **[ON-SCREEN ACTION]:** Show **Section 6.2 Confusion Matrices** grid and **Section 6.3 ROC Curves**.
- **[SPOKEN SCRIPT]:**

*"In **Sections 6.2 and 6.3**, our 2x2 confusion matrices and ROC curves demonstrate low false-negative rates, ensuring life-threatening emergency requests are not overlooked."*

● Section 7.1: Live Single-Record Prediction (2:15 – 2:35)

- **[ON-SCREEN ACTION]:** Scroll to **Section 7.1** showing the Live Text Prediction Output cell.
- **[SPOKEN SCRIPT]:**

*"In **Section 7.1**, we demonstrate live single-record prediction directly inside Colab.*

*When we pass a synthetic emergency message—'We urgent need medical assistance and clean drinking water in Les Cayes! People are injured after earthquake.'—our trained Voting Ensemble instantly predicts **DISASTER-RELATED** with **95%+ Emergency Probability**.*

*Conversely, casual text like 'Let us catch up over coffee' is correctly classified as **NON-CRISIS**."*

● Section 7.2 & 7.3: SHAP Feature Importance & Beeswarm Plot (2:35 – 2:50)

- **[ON-SCREEN ACTION]:** Point mouse at **Section 7.2 SHAP Bar Chart** and **Section 7.3 Beeswarm Plot**.
- **[SPOKEN SCRIPT]:**

"In **Sections 7.2 and 7.3**, we apply **SHAP (SHapley Additive exPlanations)** for Explainable AI.

Our SHAP feature importance plot and summary beeswarm plot confirm that key emergency terms—*water, food, shelter, help, injured, and hospitals*—have the highest positive impact on driving disaster relevance predictions."

● Section 8: Model Persistence & Pipeline Artifacts (2:50 – 3:00)

- **[ON-SCREEN ACTION]:** Scroll to **Section 8** showing the `joblib.dump` code cell.
- **[SPOKEN SCRIPT]:**

"Finally, in **Section 8**, we save our trained Voting Ensemble model weights and TF-IDF vectorizer artifacts for real-world deployment.

This completes our end-to-end Machine Learning presentation. Thank you very much!"

Part 3: Exact Numerical Verification Table

Section	Model Name	Model Type	CV F1	Test Acc	Test Prec	Test Rec	Test F1	Test ROC-AUC
Sec 5	Logistic Regression	Baseline	0.8001	78.01%	0.8729	0.7801	0.8054	0.8835
Sec 5	Random Forest	Bagging (a)	0.7582	73.26%	0.8455	0.7326	0.7649	0.8326
Sec 5	Linear SVM	Linear Classifier	0.7911	84.98%	0.8453	0.8498	0.8474	0.8745
Sec 5	XGBoost	Boosting (b)	0.7532	85.73%	0.8378	0.8573	0.8403	0.8493
Sec 5	Voting Ensemble	Soft Voting (c)	0.8105	85.65%	0.8669	0.8565	0.8609	0.8888